

PARTE 2

Condense los grados de libertad utilizando MAPLE 12 u otro programa

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Favorites
Handwriting
Expression

Units (SI)

[m]	[kg]	[s]
[N]	[J]	[K]
[T]	[A]	[V]
[C]	[Ω]	[F]
[H]	[rad]	[sr]
[mol]	[lx]	[lm]
[S]	[Wb]	[Np]

Units (FPS)

[lbf]	[ft]	[s]
[poundal]	[lb]	
[poundforce]	[HP]	
[inch ²]		
[poundal/ft]	[rad]	
[sr]		

Common Symbols

π	e	i	j	I	∞
Σ	Π	∫	d	∩	U
≥	>	≠	≤	<	
≠	≡	≠	∈	∉	⊆

Math

```
> with (LinearAlgebra) :
> Kbb := Matrix([[4*h^2, 0, 2*h^2, 0], [0, 6*h^2, h^2, 0], [2*h^2, h^2, 8*h^2, h^2], [0, 0, h^2, 6*h^2]]);
```

$$K_{bb} := \begin{bmatrix} 4h^2 & 0 & 2h^2 & 0 \\ 0 & 6h^2 & h^2 & 0 \\ 2h^2 & h^2 & 8h^2 & h^2 \\ 0 & 0 & h^2 & 6h^2 \end{bmatrix} \quad (1)$$

```
> Kbb_inv := LinearAlgebra:-MatrixInverse(Kbb);
```

$$K_{bb_inv} := \begin{bmatrix} \frac{23}{80h^2} & \frac{1}{80h^2} & -\frac{3}{40h^2} & \frac{1}{80h^2} \\ \frac{1}{80h^2} & \frac{41}{240h^2} & -\frac{1}{40h^2} & \frac{1}{240h^2} \\ -\frac{3}{40h^2} & -\frac{1}{40h^2} & \frac{3}{20h^2} & -\frac{1}{40h^2} \\ \frac{1}{80h^2} & \frac{1}{240h^2} & -\frac{1}{40h^2} & \frac{41}{240h^2} \end{bmatrix} \quad (2)$$

```
> Kba := Vector([y[1], y[2], y[3], y[4]]);
```

$$K_{ba} := \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (3)$$

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Units (SI)

[[unit]]	[[m]]	[[s]]
[[N]]	[[kg]]	[[Pa]]
[[W]]	[[J]]	[[K]]
[[T]]	[[A]]	[[V]]
[[C]]	[[Ω]]	[[F]]
[[H]]	[[rad]]	[[sr]]
[[mol]]	[[lx]]	[[lm]]
[[S]]	[[Wb]]	[[Np]]

Units (FPS)

[[unit]]	[[ft]]	[[s]]
[[poundal]]	[[lb]]	
[[poundforce]]	[[HP]]	
[[poundal.ft]]	[[rad]]	
[[sr]]		

Common Symbols

π	e	i	j	I	∞
Σ	Π	∫	d	∩	U
≥	>	≠	≠	≤	<
≠	≠	α	≈	~	=
≠	≠	≠	∈	∉	⊆

$$Kbb_inv := \begin{bmatrix} -\frac{3}{40h^2} & -\frac{1}{40h^2} & \frac{3}{20h^2} & -\frac{1}{40h^2} \\ \frac{1}{80h^2} & \frac{1}{240h^2} & -\frac{1}{40h^2} & \frac{1}{240h^2} \end{bmatrix}$$

$$Kba := \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad (3)$$

$$Kba := \begin{bmatrix} 6h \\ 6h \\ 6h \\ 6h \end{bmatrix} \quad (4)$$

$$RI := Kbb_inv.Kba;$$

$$RI := \begin{bmatrix} \frac{57}{40h} \\ \frac{39}{40h} \\ \frac{3}{20h} \\ \frac{39}{40h} \end{bmatrix} \quad (5)$$

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$R1 := Kbb_inv.Kba;$
 $R1 := \begin{bmatrix} 6h \\ \frac{57}{40h} \\ \frac{39}{40h} \\ \frac{3}{20h} \\ \frac{39}{40h} \end{bmatrix}$ (5)

$Kabx := Vector[row]([x[1], x[2], x[3], x[4]]);$
 $Kabx := \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}$ (6)

$Kab := Matrix([[6*h, 6*h, 6*h, 6*h]]);$
 $Kab := \begin{bmatrix} 6h & 6h & 6h & 6h \end{bmatrix}$ (7)

$R2 := Kab.R1;$
 $R2 = \frac{297}{20}$ (8)

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